Multi-Unit Mixing Tank, Distillation Tank and Recycle Stream Material Balance:
Project Report

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1. Problem Statement and Objective:

Two ethanol-water streams are fed into a continuous steady-state multi-unit system consisting of a mixing tank, distillation column and a splitter. In our problem, Feed 1 is fixed at 100 kg/h and Feed 2 is 50 kg/h (base case); the composition is 10 wt% ethanol in Feed 1 and 60 wt% ethanol in Feed 2, and the mixer outlet is fed to the distillation column. The column is specified to recover 95% of the ethanol in its feed in the distillate, with a distillate composition of 80 wt% ethanol. 50% of the bottoms stream is recycled back to the mixing tank, and the remaining is Purge.

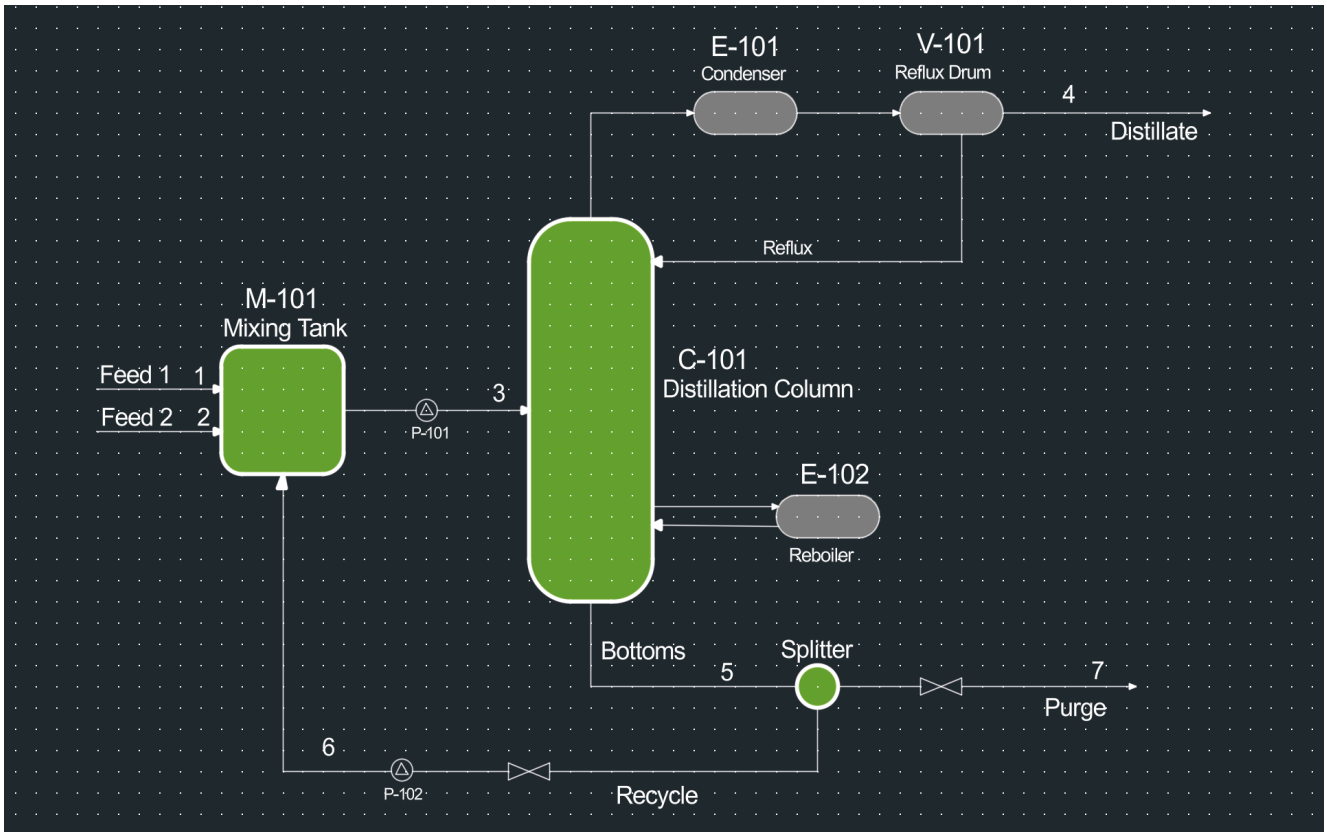
The objectives are to calculate the flow rate and composition of every stream, with the products and purge as outlets, find the feed 2 flow rate needed to reach a target ethanol composition of 20 wt% in the column feed. And lastly, test how sensitive ethanol recovery, recycle flow, column feed composition and product flow are to each input using Python.

2. Assumptions:

- Steady state, so there is no accumulation in any unit
- No chemical reaction
- Perfect mixing

- Compositions are all on a mass basis (Volumes are not additive)
- The column is only modelled by its recovery and purity specs, not step-by-step design
- The splitter produces two streams of identical composition

3. Process Diagram:



The greens units form our system.

Stream 1:- Fresh feed 1

2:- Fresh feed 2

3:- Mixer Outlet

4:- Distillate(product)

5:- Column Bottoms

6:- Recycle Stream

7:- Purge

4. Given Data:

Item	Value
Feed 1 flow rate	100 kg/h
Feed 1 Ethanol , x (mass fraction)	0.10
Feed 2 flow rate (base case)	50 kg/h
Feed 2 Ethanol, x	0.60
Ethanol Recovery in column, p	0.95
Distillate ethanol, x	0.80
Bottoms recycled, r (recycle fraction) (base case)	0.50
Target ethanol in column feed(stream 3)	20 wt%

5. Degrees of freedom analysis :

The Unknowns are flow rates at streams 3 to 7 (5) and both ethanol and water compositions in streams 3,5,6 and 7 (4)
Implies No. of Unknowns = 9
No. of Equations include:
Mixer mass and component balance: 2
Distillation column mass and component balance: 2
Splitter equations : 3
Problem Specifications: 2
DOF = No. of unknowns - No. of Independent Equations
= 9 - (2+2+3+2) = 9 - 9 = 0
Hence, the problem is precisely specified (sufficient data given) and is perfectly solvable.

6. Equations and Hand Calculations :

Symbols ‘m’ used for mass flow rates and ‘x’ used for mass fraction

Overall System Material Balance:

$m_1 + m_2 = m_4 + m_7$

Overall System Component Balance:

Etoh: $x_E^1 m_1 + x_E^2 m_2 = x_E^4 m_4 + x_E^7 m_7$

Water: $x_W^1 m_1 + x_W^2 m_2 = x_W^4 m_4 + x_W^7 m_7$

Mixer System Material Balance:

$m_1 + m_2 + m_6 = m_3$

Mixer component Material Balance:

Etoh: $x_E^1 m_1 + x_E^2 m_2 + x_E^6 m_6 = x_E^3 m_3$

Water: $x_W^1 m_1 + x_W^2 m_2 + x_W^6 m_6 = x_W^3 m_3$

Distillate column System Material Balance:

$m_3 = m_4 + m_5$

Distillate column component Material Balance:

Etoh: $x_E^3 m_3 = x_E^4 m_4 + x_E^5 m_5$

Water: $x_W^3 m_3 = x_W^4 m_4 + x_W^5 m_5$

Splitter Equation:

$x_E^5 = x_E^6$

$x_E^5 = x_E^7$

(and so are the compositions of water, which is basically 1 - x_E)

Process Specification Equations:

$x_E^4 m_4 = (.95)(x_E^3 m_3)$ → Recovery specs

$m_6 = 0.50(m_5)$ or $m_7 = 0.50(m_5)$ → Recycle fraction

On substituting the given values and solving these equations, we get the following values:

E_X is the flow rate of ethanol in x stream, and W_X is the flow rate of water in x stream

$E_3 = E_{1+2} /(1\text{-}r(1\text{-}p))$ from the spec equations; 95% goes to distillate, and 5% goes to bottoms; half of the bottoms is recycled; it brings r(1-p) ethanol to the stream

$W_3 = (W_{1+2} - r*W_4)/(1\text{-}r)$ water leaving distillate is fixed by 80% purity spec: bottoms water is $W_3 - W_4$; recycle returns r of it

Base-Case Results:

Step	Result
Fresh feed	E_{1+2}= 40 kg/h , W_{1+2}= 110 kg/h
Column feed ethanol	E_3 = 40/(1-0.5x0,05) = 41.026 kg/h
Distillate ethanol	E_4 =0.95 x 41.026 = 38.974 kg/h
Bottoms ethanol	E_5 = 41.026 - 38.974 = 2.051 kg/h
Distillate water	W_4 =38.974 x 0.20/0.80 = 9.744 kg/h
Column feed water	W_3 =(110 - 0.5x9.744)/0.5 = 210.256 kg/h
Bottoms water	W_5 = 210.256 - 9.744 = 200.513 kg/h
Recycle	E_6 =1.026 kg/h, W_6 = 100.256 kg/h
Purge	E_7 =1.026 kg/h, W_7 = 100.256 kg/h

The Python code has identical results to this table; the overall balance closes to within numerical precision (residuals below 10^{-13} kg/h), considering the accuracy up to 3 decimal places.

7. Feed 2 Flow required for a Target Column-Feed Composition (Target case):-

To solve the second part of our objective we are required to find feed 2 flow rate required to achieve 20 wt% composition. Now for reaching a chosen mass fraction z of ethanol in stream3 at fixed r and p, the feed 2 flow F_2 is found by solving the two column-feed equations above for F_2:
Writing a=1/[1-r(1-p)] , q=(1-z)/z
And k =$(1-x_E^4) / x_E^4$ the condition $W_3 = q\ E_3$ reduces to:

$$W_{1+2} = c\ .\ E_{1+2}\ \text{, where } c = a[(1 - r)q + r \cdot k \cdot p]$$

Because E_{1+2} and W_{1+2} are both linear in F_2 , this gives a direct solution:

$$F_2 = F_1\ [(1 - x_1) - c \cdot x_1] / [c \cdot x_2 - (1 - x_2)]$$

For z = 0.20: a = 1.02564, q = 4, k = 0.25,
so c = 1.02564 × (0.5 × 4 + 0.5 × 0.25 × 0.95) = 2.17308, and

$$F_2 = 100\ (0.90 - 0.21731) / (1.30385 - 0.40) = 75.53 \text{ kg/h}$$

This is 51% more than the base-case 50 kg/h, which gives 16.33 wt% in stream 3. The function f2_for_target in flowsheet.py implements this formula and raises an error if the target cannot be reached with the given x2 , r and p, or if it is below what Feed 1 alone already gives. The resulting streams are:

Target Case:
Results:

```
F2 for 20% ethanol in stream 3: 75.53 kg/h
stream 3 check: matrix 20.00%, iteration 20.00%

stream   ethanol    water    total   %EtOH   (target case)
    s3    56.738  226.950  283.688   20.00
    s4    53.901   13.475   67.376   80.00
    s5     2.837  213.475  216.312    1.31
    s6     1.418  106.738  108.156    1.31
    s7     1.418  106.738  108.156    1.31
```

8. Code Overview:

The Python code solves the same balances in two independent ways. The first writes all 10 equations as rows of a matrix and uses NumPy to find the one set of 10 unknowns that satisfies every row at the same time. The code uses component flows while the Dof analysis uses total flows, so both describe the same problem. The second is an iterative loop that starts with a recycle guess of zero, runs the mixer, column and splitter in order, and repeats with the new recycle until it stops changing. Both give the same answer, which acts as a solution check. The code itself consists of 3 parts:

- unit.py file that holds the code for each subsystem's material balances by defining a function on each one. The mixer consists of the basic material equation that adds up the feeds to get the outlet stream flow. The column holds the spec equations used to divide the inlet flow into bottoms and distillate; it also raises an error if specs need more water than what the feed contains. The splitter divides the bottoms stream into two streams of the same composition by making use of the recycle fraction
- flowsheet.py is the file that holds the problem solver, i.e the matrix solver that solves the equations altogether, and the iterative method that runs 40 iterations in our case, starting with recycle of 0 and

stopping when recycle reaches a constant. It also holds the code for finding flow rate of feed 2 required for target composition.

- run.py is the file that runs the solver with both methods, confirms that both methods agree and prints a table consisting of all missing values. It also checks for the overall material balance as a re-verification for the right values (just like we do). It calculates the ethanol recovery and how sensitive the recovery is with respect to the recycle flow change.

9. Result analysis:

```
stream   ethanol    water    total   %EtOH
    s3    41.026  210.256  251.282   16.33
    s4    38.974    9.744   48.718   80.00
    s5     2.051  200.513  202.564    1.01
    s6     1.026  100.256  101.282    1.01
    s7     1.026  100.256  101.282    1.01

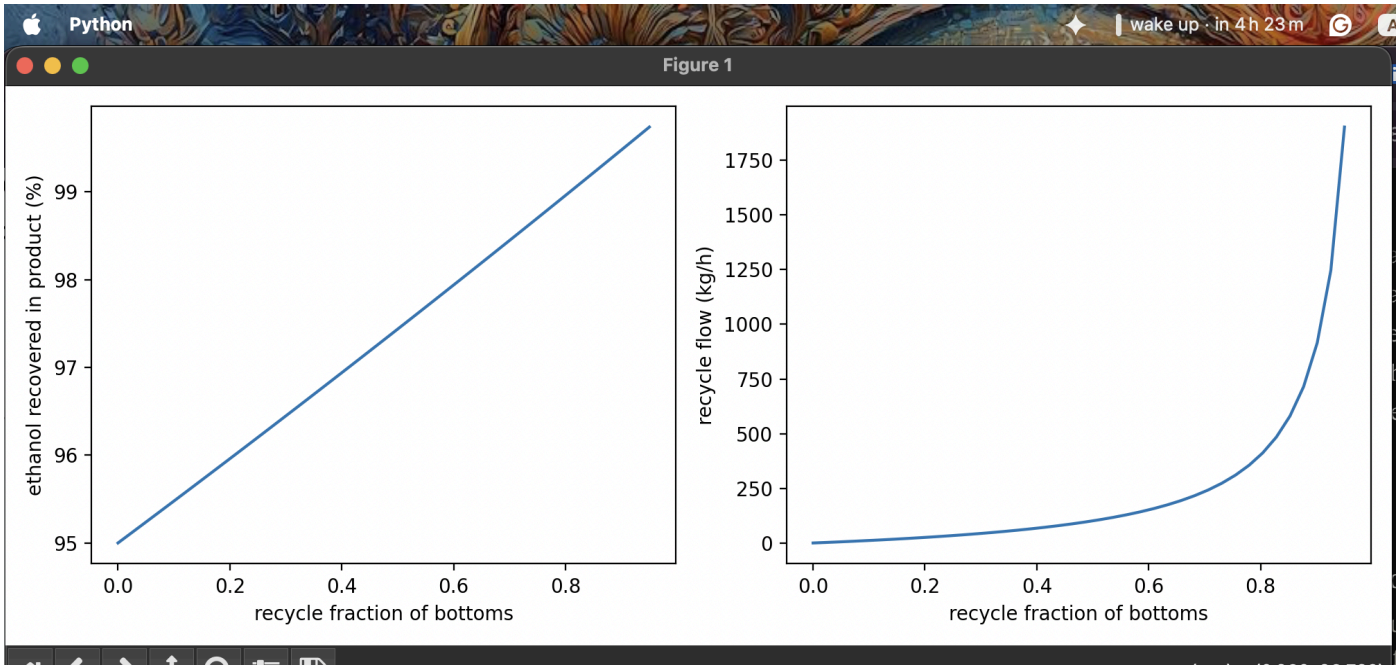
matrix and iteration agree: True (40 iterations)
overall balance residual: [ 3.33066907e-15 -2.84217094e-14]
ethanol recovered in product: 97.4%
```

These results from Python, as stated above, match the hand calculations to a very high accuracy; as mentioned, the result displays whether the two methods agree and how many iterations it took for method 2 to arrive at the final solution.

A) Recycle Fraction sweep:

Both these graphs reflect a parametric sweep on recycle fractions. It puts in one input across a range (0 to 95%) and watches how the output

reacts. Here we observe changes in recycle flow and ethanol recovery with respect to recycle fraction :

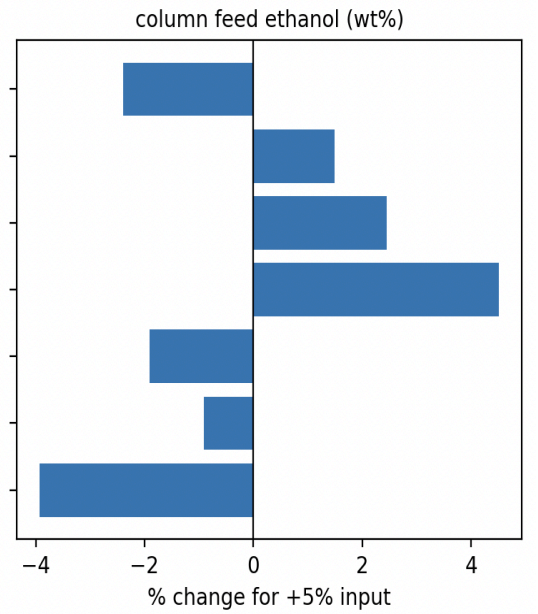


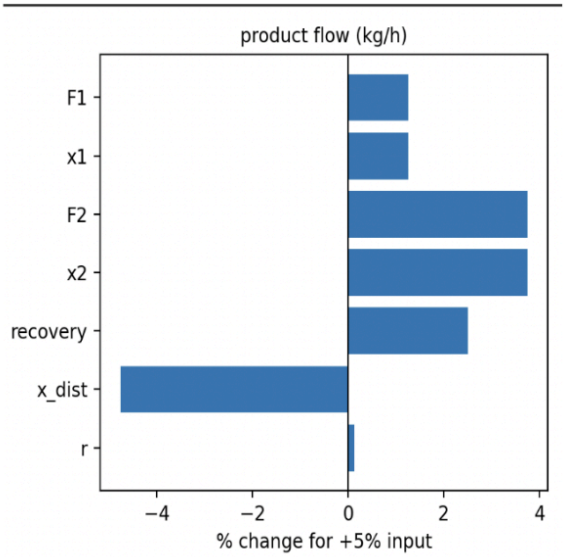
- Recycle fractions vs ethanol recovery: We observe that ethanol recovery rises from 95% at 0 recycle to about 99.7% at a recycle fraction of 0.95 since it returns the ethanol that would otherwise leave the system in purge.Algebraically it is $p / [1 - r(1 - p)]$, which is nearly linear over this range. The recovery is only about 4.7 percentage points over the whole range.
- Recycle fractions vs recycle flow: the graph is nearly flat at first and then rises steeply, as r almost equals 1, at high recycle fractions. The recycle is mostly water, and the water leaving in the purge is fixed by the overall balance, so this smaller purge fraction forces the water to build up in the loop, thus causing such high flow rates.

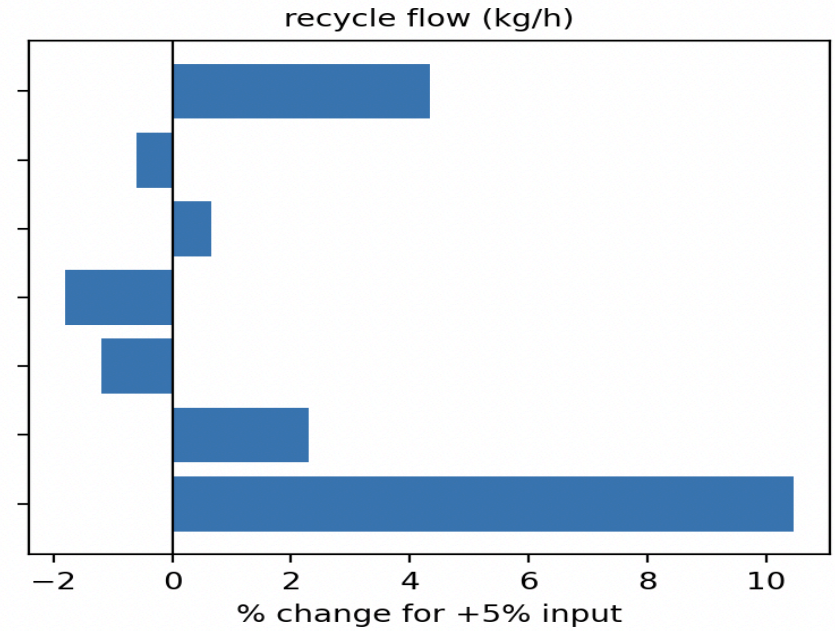
B) Sensitivity to the inputs:

These are sensitivity graphs that change the inputs by 5%; here we compare several inputs like r, x_{dist}, F_1, F_2, recovery, x_1 and x_2 to see how four outputs: product and recycle flow, column feed ethanol, and ethanol recovery are affected by them. Based on the three graphs:

Input (+5%)	Column feed ethanol (wt%)	Product flow (kg/h)	Overall ethanol recovery (%)	Recycle flow (kg/h)
F_1	−2.40	1.25	0.00	4.34
x_1	1.50	1.25	0.00	−0.60
F_2	2.46	3.75	0.00	0.66
x_2	4.51	3.75	0.00	−1.80
recovery (p)	−1.90	2.50	2.50	−1.20
x_{dist}	−0.91	−4.76		2.29
r	−3.92	0.13	0.13	10.46


column feed ethanol (wt%)
−4
−2
0
2
4
% change for +5% input


product flow (kg/h)
F1
x1
F2
x2
recovery
x_dist
r
−4
−2
0
2
4
% change for +5% input


recycle flow (kg/h)
−2
0
2
4
6
8
10
% change for +5% input

- Product flow depends highly on distillate purity(−4.8%) then by feed 2 flow and composition (+3.75% each) and then column recovery(+2.5%), followed by feed 1 that give +1.25% each and almost nothing by the recycle fraction. This implies that product flow depends on how much ethanol is fed, mainly through the ethanol-rich stream(feed 2) and distillate purity <0%. This implies that column feed depends heavily on the quality of feed 2 and on the recycle fraction because a purer distillate carries less water, and hence the recycle feed, which is mostly water, affects it the most.
- The recycle flow, as we saw in the sweep graphs, is affected heavily by recycle fraction (+10.5%), followed by feed 1 flow (+4.3%), distillate purity (+2.3%), and feed 2 composition (−1.8%). The remaining 3 are below are less than 1.2% and hence matter less. This implies that recycle flow, which decides the pump and pipe sizes, is controlled mainly by the recycle fraction.
- The column feed is also largely affected by feed 2 composition (+4.5%) , then the recycle fraction (−3.9%). Feed 2 flow (+2.5%), feed 1 flow (−2.4%), column recovery (−1.9%), feed 1 composition (+1.5%), and distillate purity (−0.9%, since a purer distillate carries away less water, leaving more in the loop) have smaller effects.
- Overall ethanol Recovery isn't plotted as it responds only to column recovery (+2.5%) and recycle fraction(+0.13%), and <0% for the others implies that it mostly depends on the column recovery.

10. Real World Implications:

Although the problem is a very textbook case, the sensitivity graphs help us evaluate the efficiency of such a system in these conditions and help us to identify flaws such as what works and what doesn't for the system. Here we noticed from the graphs that increasing the fraction of bottoms recycled ultimately caused the recovery of ethanol to increase well from 95% to 97.4% . This is the separation efficiency of the whole plant. The only real benefit of this recycle stream is that less ethanol leaves in the purge, but the recovery gain is very small.

Relating this to the second graph shows that because the recycle stream consists mainly of water, it causes the flow rate to increase, thus forcing a need to increase the diameters of pipes to run flow comfortably. But that itself is a cost-intensive process for a very small increase in recovery. Operating costs for heating the extra water would also rise. All this increase in cost for a very small gain is highly impractical. Thus, a real-life plant would be less likely to recycle a stream that is mostly water just to recover a trace of ethanol.

Limitations of this system is that the column is modelled only by its recovery and purity specifications, so energy use, the number of stages and the vapour-liquid equilibrium are not checked. The 80 wt% distillate is below the ethanol-water azeotrope (about 95.6 wt%) , so specification is physically reachable by ordinary distillation.